- Close the key and hence the circuit is closed.
- Adjust the rheostat such that the ammeter reads a current of 0.5 ampere.
- Note down the potential difference across the resistor as shown by the voltmeter.
- Adjust the rheostat and change the current in steps of 0.5A (that is 0.5A, 1.0A, 1.5A, etc.).
- For each current, note down the corresponding potential difference as shown by the voltmeter.
- Tabulate the observations.
- Measure the diameter of the wire using a screw gauge.
- Measure the length of the coil using metre scale

Observations:

(i) To find the resistance:

S. No	Ammeter reading-I (Ampere)	Voltmeter reading-V (Volt)	Resistance = V/I (Ohm)
1			
2			
3			
		MEAN	

(ii) To find the diameter of the wire using screw gauge:

S. No	Pitch Scale reading-PSR (mm)	Head scale coincidence-HSC	Head scale reading-HSR=HSC×LC (mm)	Total reading = PSR + HSR (mm)
1				
2				
3				
			Mean Diameter	

Calculations:

Radius of the wire, r = diameter/2 = _____ m

Area of cross section of the wire, A = πr^2 = _____ m^2

Length of the wire L = _____ m.

Resistivity of the material of the wire = $\rho = \left(\frac{A}{L}\right)R$ = _____ Ω m

Result:

The resistivity of the material of the wire = _____ Ω m

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